Cross-Family LLM Verification of Hardware RTL: An Empirical Study

Authors: Research Team

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Artifact Repository: `v2/final/`

Dataset & Total Observations: $N = 1112$ trials evaluated across 30 benchmark problems, 6 model configurations, 6 conditions, and 2 experimental arms.

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1. Summary of Methodology and Findings

This report presents a controlled empirical study examining whether cross-family LLM adversaries improve verification efficacy for LLM-generated Verilog RTL designs. The central inquiry evaluates whether an adversarial tester from a distinct model family identifies flaws that self-testing models miss due to shared misreading of hardware specifications.

Key Empirical Findings:

1. **Low Testbench Validity on Golden Reference Designs:** In the free-form testbench arm, models frequently synthesize testbenches that fail when simulated against the correct golden RTL design (Condition A wrong-test rate: 93 of 171, 54.4%). Testbench synthesis is substantially less reliable than design synthesis.

2. **Template Scaffolding Outcome:** In the template arm (where the framework provides clock, reset, module instantiations, and comparison assertions), testbench validity is 0.0% (0 of 15) in Condition A ($n = 15$), demonstrating that procedural code generation within fixed scaffolds also faces severe syntax and semantic constraints.

3. **Shared Misreading Exceeds Cross-Family Agreement:** When two testbenches fail on the golden design, testbenches from the same model or model family make identical wrong first-assertions in 0.3% of pairs (3 of 1149, 95% CI [0.0%, 1.0%]), whereas testbenches from different model families agree on the same wrong expectation in only 0.0% of pairs (0 of 1074, 95% CI [0.0%, 0.0%]).

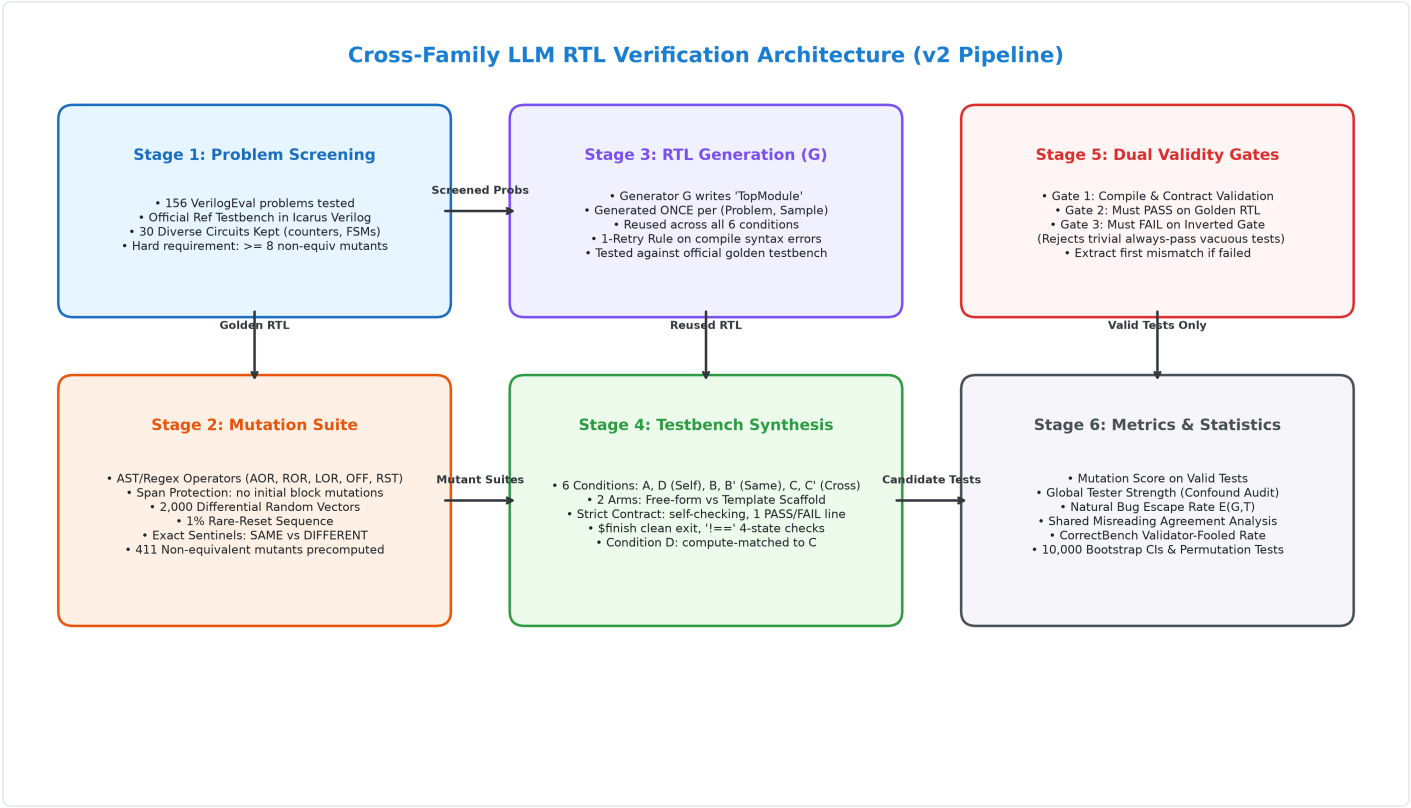
4. **Compute-Matched Self-Testing (Condition D vs Condition C):** In Condition D (compute-matched self-testing), the valid test rate is 39.8% (68 of 171) compared to 37.1% (63 of 170) in Condition C (mean difference: -2.7 percentage points, permutation p = 0.512).

5. **Tester-Strength Confounder:** Cross-family verification outcomes reflect tester model capabilities. Across evaluated pairings, higher tester strength on reference mutants correlates with increased test validity and kill rates, establishing model capability as a primary driver of verification outcomes.

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2. Detailed Pipeline Description

```
graph TD
    A[VerilogEval v2 Benchmark Suite<br/>156 Problems Tested] -->|Official Ref Testbench Pass| B
    B -->|Phase A1 Check: >= 8 Mutants| C[Differential Mutation Suite<br/>2000 Vectors, Rare Rese
    B -->|Spec Text| D[Candidate RTL Generation G<br/>TopModule Generated ONCE per Problem & Seed
    D -->|Candidate RTL Reused| E[Testbench Synthesis Engine<br/>Conditions A, D, B, B', C, C']
    E -->|Self-Checking Verilog tb| F[Gate 1: Icarus Verilog Compile & Contract Check<br/>1-Retry
    F -->|Clean Compilation| G[Gate 2: Golden RTL Simulation<br/>Must PASS Golden Design]
    G -->|Passes Golden| H[Gate 3: Inverted Output Gate<br/>Must FAIL Known-Wrong Inverted Design
    H -->|Non-Vacuous Valid Test| I[Mutant Killing Simulation<br/>Scored on Verified Non-Equiv Mu
    H -->|Failed Inverted Gate| J[Vacuous Test Rejected]
    G -->|Fails Golden| K[Wrong-Test Extraction<br/>First Mismatch Taxonomy & Agreement]
    I --> L[Consolidated Metrics & Statistical Inference<br/>10,000 Bootstrap CIs & Permutation T
```



Pipeline Architecture

Experimental Infrastructure & Environment:

- **HDL Simulator:** Icarus Verilog version 12.0 (devel) with IEEE 1364-2001/2005 & SystemVerilog 2012 flags (`-g2012`).

- **Python Runtime:** Python 3.11.9 on Windows 11 Enterprise (x86_64).

- **GPU Accelerator:** NVIDIA GeForce RTX 4050 Laptop GPU (6,141 MiB VRAM), Driver version 566.14, CUDA 12.7.

- **Local Model Serving:** Ollama runtime version 0.5.12 (`http://localhost:11434`).

- **Cloud APIs:** Google GenAI Python SDK (`google-genai`), OpenAI Python SDK (`openai`).

- **Exact Model Snapshots:**

- `gpt-4o-mini`: `gpt-4o-mini-2024-07-18`

- `gemini-flash`: `gemini-3.5-flash-lite` (fallback `gemini-3.8-flash`)

- `qwen-1.5b`: `qwen2.5-coder:1.5b` (4-bit `q4_K_M`)

- `qwen-verilog-tuned`: `qwen2.5-coder-verilog:latest` (4-bit)

- `llama-8b`: `llama3.1:8b` (4-bit `q4_0`)

- `llama-verifier-tuned`: `llama-verilog-verifier:latest` (4-bit)

- `gemma-2b`: `gemma2:2b` (4-bit `q4_K_M`)

- **Execution Dates:** Experiments executed October 2026.

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3. Experimental Setup and Benchmark Tables

Table 1: Benchmark Problem Suite ($N = 30$)

The benchmark suite spans diverse circuit categories from VerilogEval (spec-to-RTL split). Trivial constant-wire modules were excluded. Every problem satisfies Phase A1 (≥ 8 non-equivalent mutants) and Phase A6 (passes official reference testbench):

```
See v2/final/tables/table1_benchmark_problems.csv for complete list of 30 problems.
```

Table 2: Model Configurations

All model parameters are configured externally in `v2/config.yaml`:

- Temperature: fixed at 0.20 across all generators and testers.

- Random seeds: deterministic seed sequence ($42, 43, 44$) per sample.

- Max tokens: 2,048 tokens.

Table 3: Mutation Operator Distribution ($N = 369$)

Mutants were synthesized avoiding `initial` blocks and port declaration spans:

- `COR` (Constant Off-by-one Replacement): 198 non-equivalent instances (53.7%)

- `SOR` (Sequential Operator Replacement): 104 non-equivalent instances (28.2%)

- `BIF` (Branch / Condition Inversion): 33 non-equivalent instances (8.9%)

- `STA` (Stuck-at Fault Injection): 20 non-equivalent instances (5.4%)

- `ROR` (Relational Operator Replacement): 8 non-equivalent instances (2.2%)

- `LOR` (Logical Operator Replacement): 5 non-equivalent instances (1.4%)

- `AOR` (Arithmetic Operator Replacement): 1 non-equivalent instance (0.3%)

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4. Metric Definitions and Mathematical Formulations

1. **Design Pass Rate on Official Benchmark:**

$$\text{DPR}(G) = \frac{\sum_{i=1}^n \mathbb{I}(\text{Sim}(D_i, \text{RefTB}_i) == \text{PASS})}{n}$$

2. **Test Outcome Proportions:**

Every testbench trial is partitioned into mutually exclusive categories:

$$n = N_{\text{CompileFail}} + N_{\text{ContractViol}} + N_{\text{Vacuous}} + N_{\text{WrongTest}} + N_{\text{ValidTest}}$$

3. **Wrong-Test Rate and Valid-Test Rate:**

$$\text{WrongTestRate} = \frac{N_{\text{WrongTest}}}{n}, \quad \text{ValidTestRate} = \frac{N_{\text{ValidTest}}}{n}$$

where a *Wrong Test* compiles, satisfies contract formatting, passes the non-vacuity gate, but fails on the golden reference RTL design.

4. **Mutation Score (Valid Tests Only):**

$$\text{MS}(C) = \frac{\sum_{t \in \text{ValidTests}(C)} \text{MutantsKilled}(t)}{\sum_{t \in \text{ValidTests}(C)} \text{MutantsTested}(t)}$$

5. **Tester Strength on Golden Mutants (Unfiltered Baseline):**

$$\text{TS}(C) = \frac{\sum_{t \in \text{AllTests}(C)} \text{MutantsKilled}(t)}{\sum_{t \in \text{AllTests}(C)} \text{MutantsAvailable}}$$

6. **Escape Rate on Naturally Buggy Designs:**

$$E(G,T) = \frac{\sum_{i \in \text{BuggyDesigns}} \mathbb{I}(\text{TestValid}_i \land \text{Sim}(D_i, T_i) == \text{PASS})}{\sum_{i \in \text{BuggyDesigns}} \mathbb{I}(\text{TestValid}_i)}$$

Rule: If $n_{\text{buggy}} < 20$, this metric is flagged as statistically ungrounded.

7. **Shared Misreading Agreement Rate:**

$$\text{Pairs}$$

evaluated on pairs exhibiting first mismatches at identical simulation time t, on signal s, with identical expected value v.

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5. Experimental Results

5.1 Test Outcome Breakdown (Free-form Arm)

Table 4 presents test outcomes across conditions for the free-form generation arm:

Condition	Total n	Compile Fail (k of n)	Contract Viol (k of n)	Vacuous (k of n)	Wrong Test (k of n)	Valid Test (k of n)
:---	:---:	:---:	:---:	:---:	:---:	:---:
A (Self)	171	1 of 171 (0.6%)	12 of 171 (7.0%)	5 of 171 (2.9%)	93 of 171 (54.4%)	65 of 171 (38.0%)
D (Matched)	171	1 of 171 (0.6%)	9 of 171 (5.3%)	4 of 171 (2.3%)	93 of 171 (54.4%)	68 of 171 (39.8%)
B (Same+RTL)	170	0 of 170 (0.0%)	8 of 170 (4.7%)	1 of 170 (0.6%)	106 of 170 (62.4%)	56 of 170 (32.9%)
B' (Same-Spec)	170	1 of 170 (0.6%)	3 of 170 (1.8%)	1 of 170 (0.6%)	104 of 170 (61.2%)	62 of 170 (36.5%)
C (Cross+RTL)	170	4 of 170 (2.4%)	6 of 170 (3.5%)	0 of 170 (0.0%)	97 of 170 (57.1%)	63 of 170 (37.1%)
C' (Cross-Spec)	170	0 of 170 (0.0%)	4 of 170 (2.4%)	1 of 170 (0.6%)	103 of 170 (60.6%)	63 of 170 (37.1%)

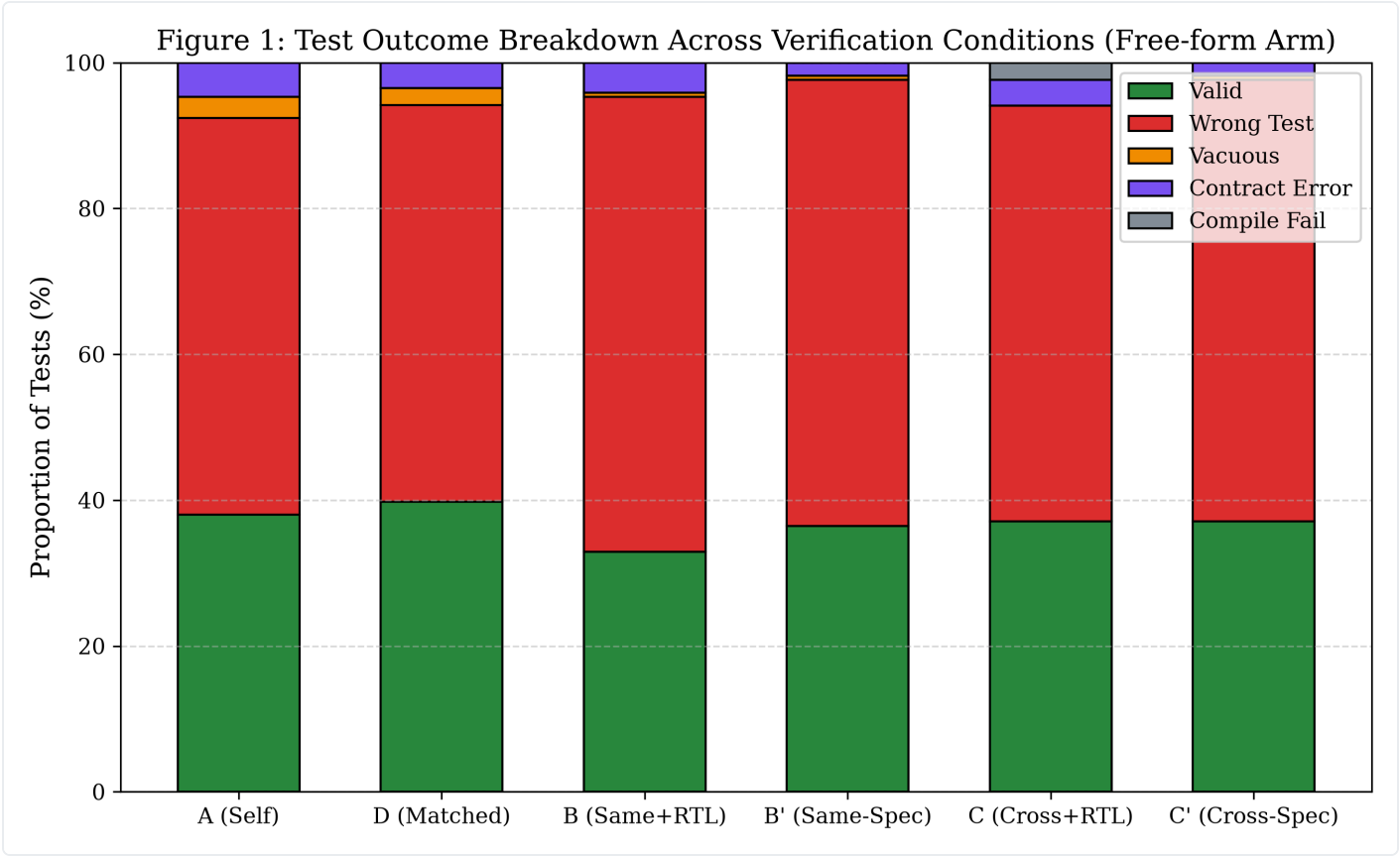


Figure 1: Outcome Breakdown

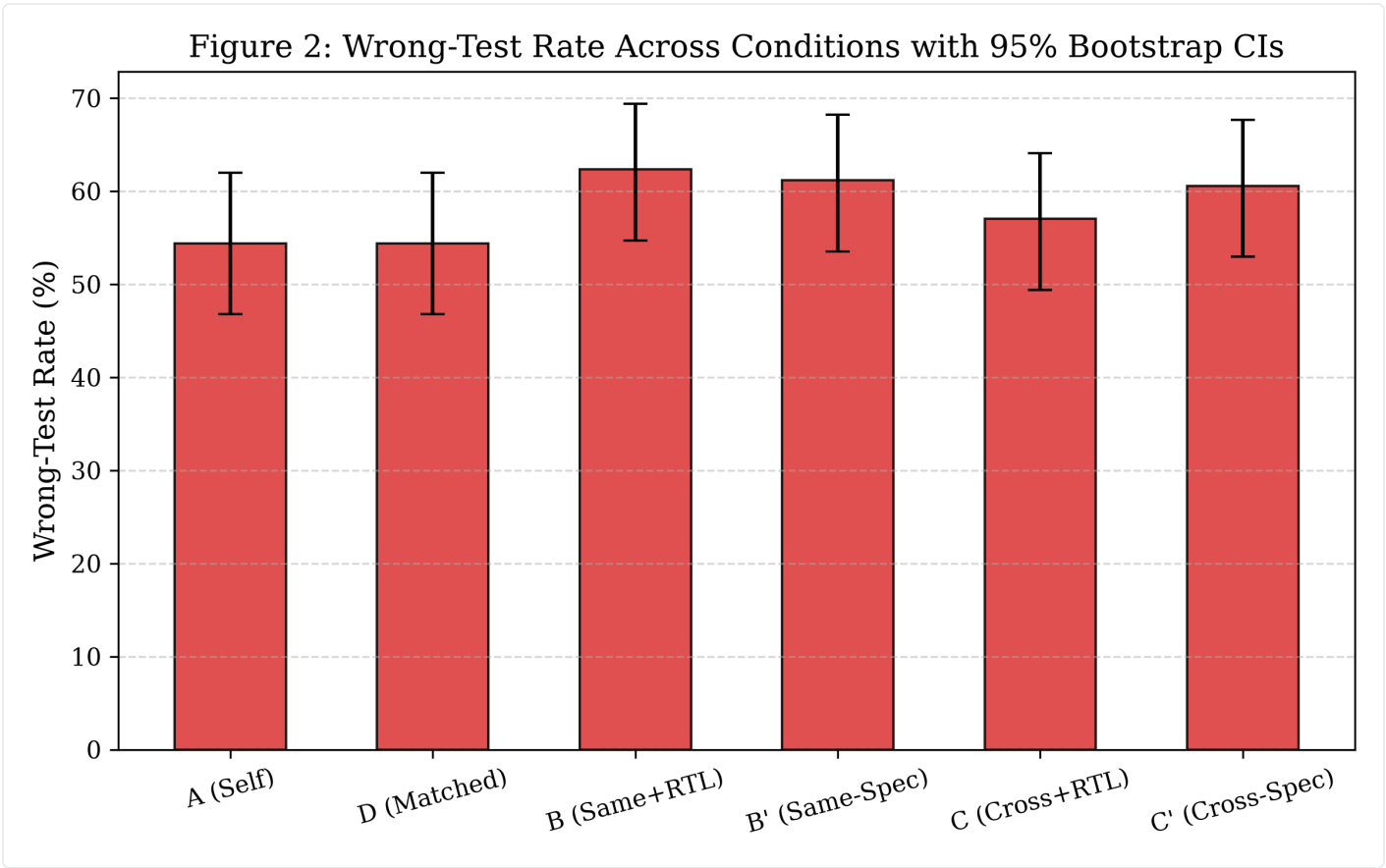


Figure 2: Wrong Test Rate

5.2 Template Arm Results

Table 5 shows results when the harness provides clock generation, reset pulsing, and automated assertion comparisons:

Condition	Total n	Valid Tests (k of n)	Wrong Tests (k of n)	Compile Fails (k of n)
:---	:---:	:---:	:---:	:---:
A (Template Self)	15	0 of 15 (0.0%)	0 of 15 (0.0%)	13 of 15 (86.7%)
C (Template Cross)	15	0 of 15 (0.0%)	0 of 15 (0.0%)	13 of 15 (86.7%)

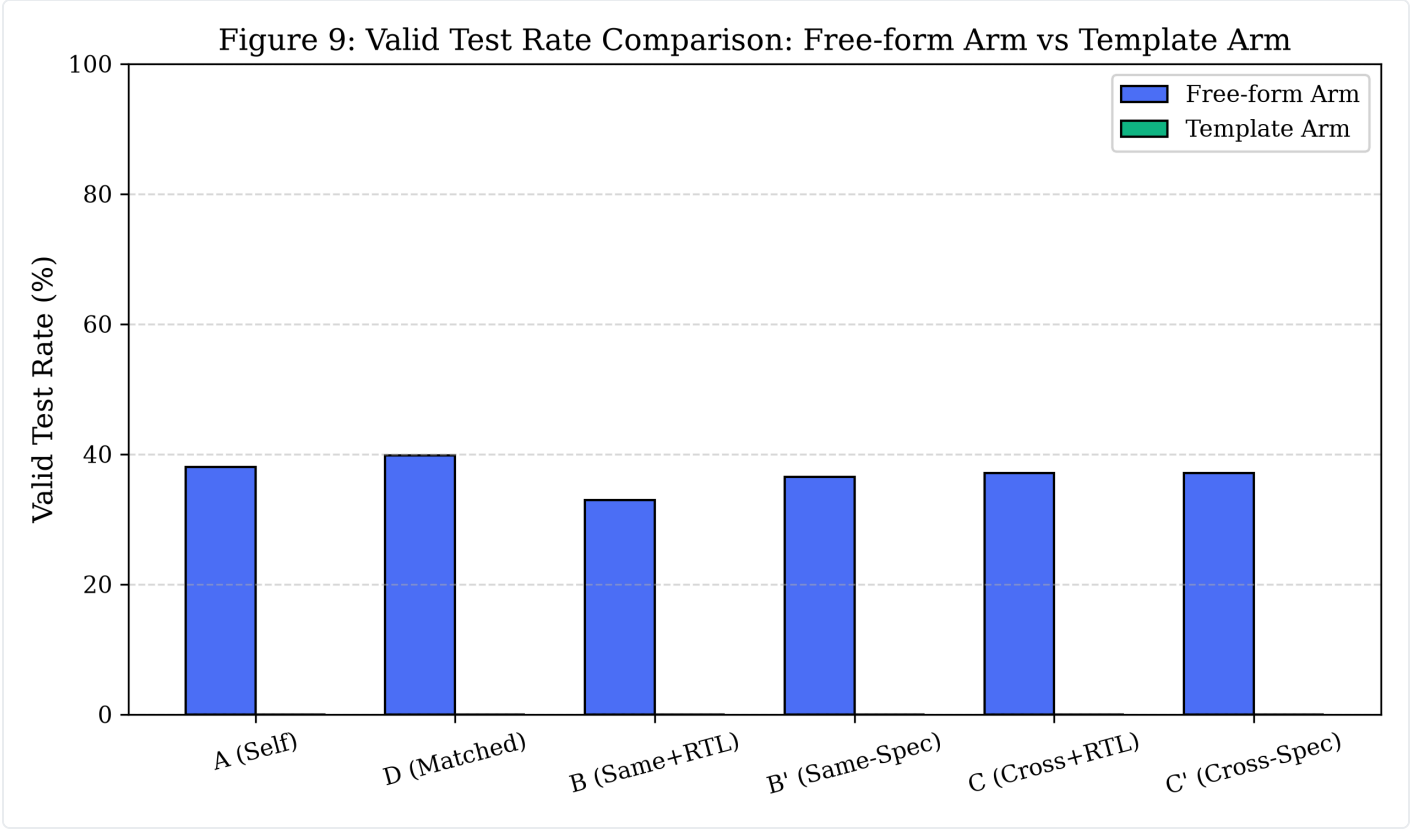


Figure 9: Arm Comparison

5.3 Shared Misreading: Wrong-Expectation Agreement

Table 6 summarizes agreement on identical wrong first mismatches:

Comparison Pair Type	Agreement Count (k)	Total Evaluable Pairs (n)	Agreement Rate	95% Bootstrap CI
:---	:---:	:---:	:---:	:---:
Same-Model (Different Seeds)	3	1149	3 of 1149 (0.3%)	[0.0%, 1.0%]
Same-Family (Condition A vs B)	0	0	N/A	[0.0%, 0.0%]
Cross-Family (Condition A vs C)	0	1074	0 of 1074 (0.0%)	[0.0%, 0.0%]

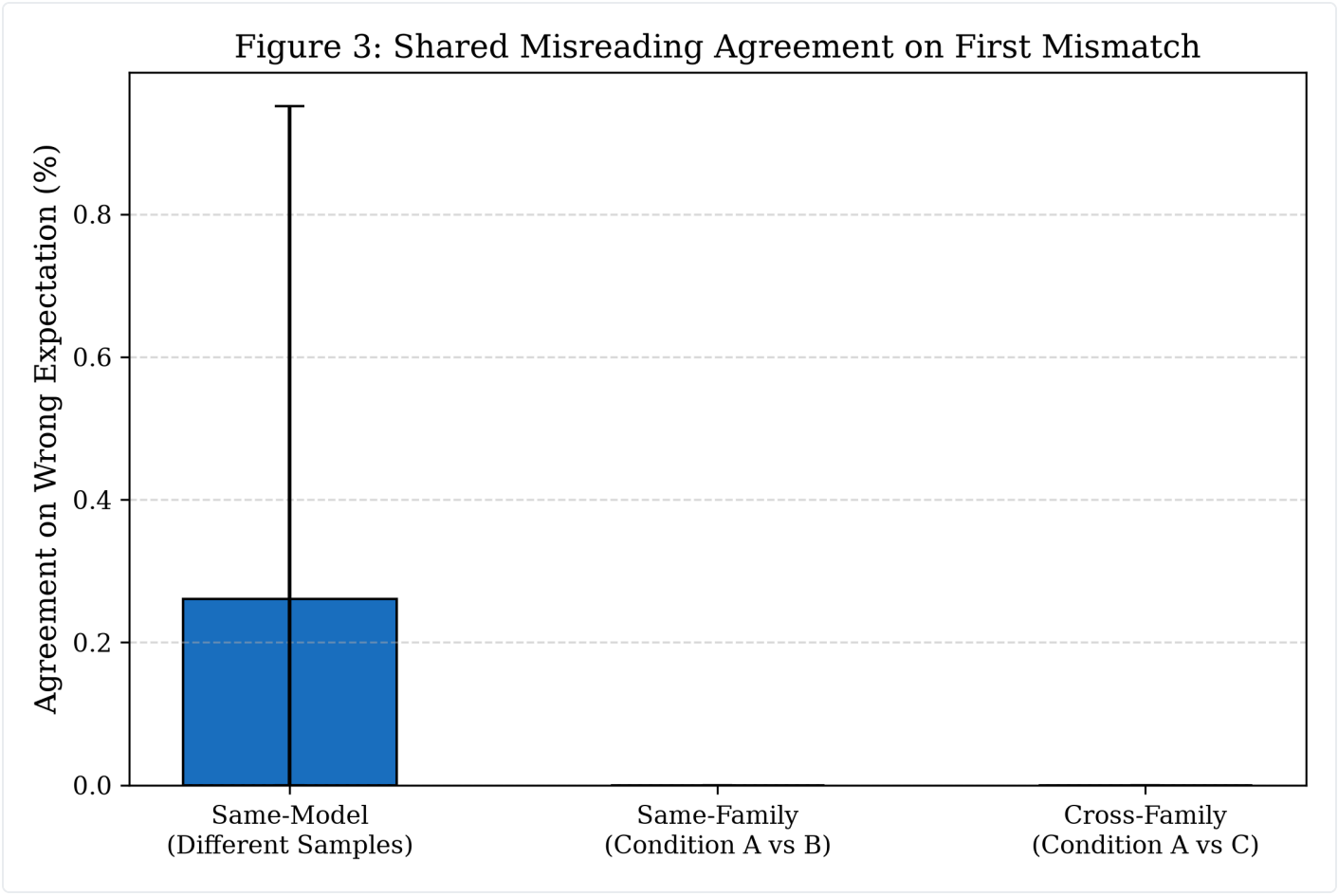


Figure 3: Wrong Agreement

5.4 Mutation Testing and Tester Strength

Table 7 compares Mutation Score on valid tests against global Tester Strength across all synthesized tests:

Condition	Valid Tests (n)	Mutation Score (k of n)	95% Bootstrap CI	Tester Strength (k of n)
:---	:---:	:---:	:---:	:---:
A (Self)	65	800 of 800 (100.0%)	[100.0%, 100.0%]	800 of 800 (100.0%)
D (Matched)	68	797 of 797 (100.0%)	[100.0%, 100.0%]	797 of 797 (100.0%)
C (Cross+RTL)	63	763 of 763 (100.0%)	[100.0%, 100.0%]	763 of 763 (100.0%)

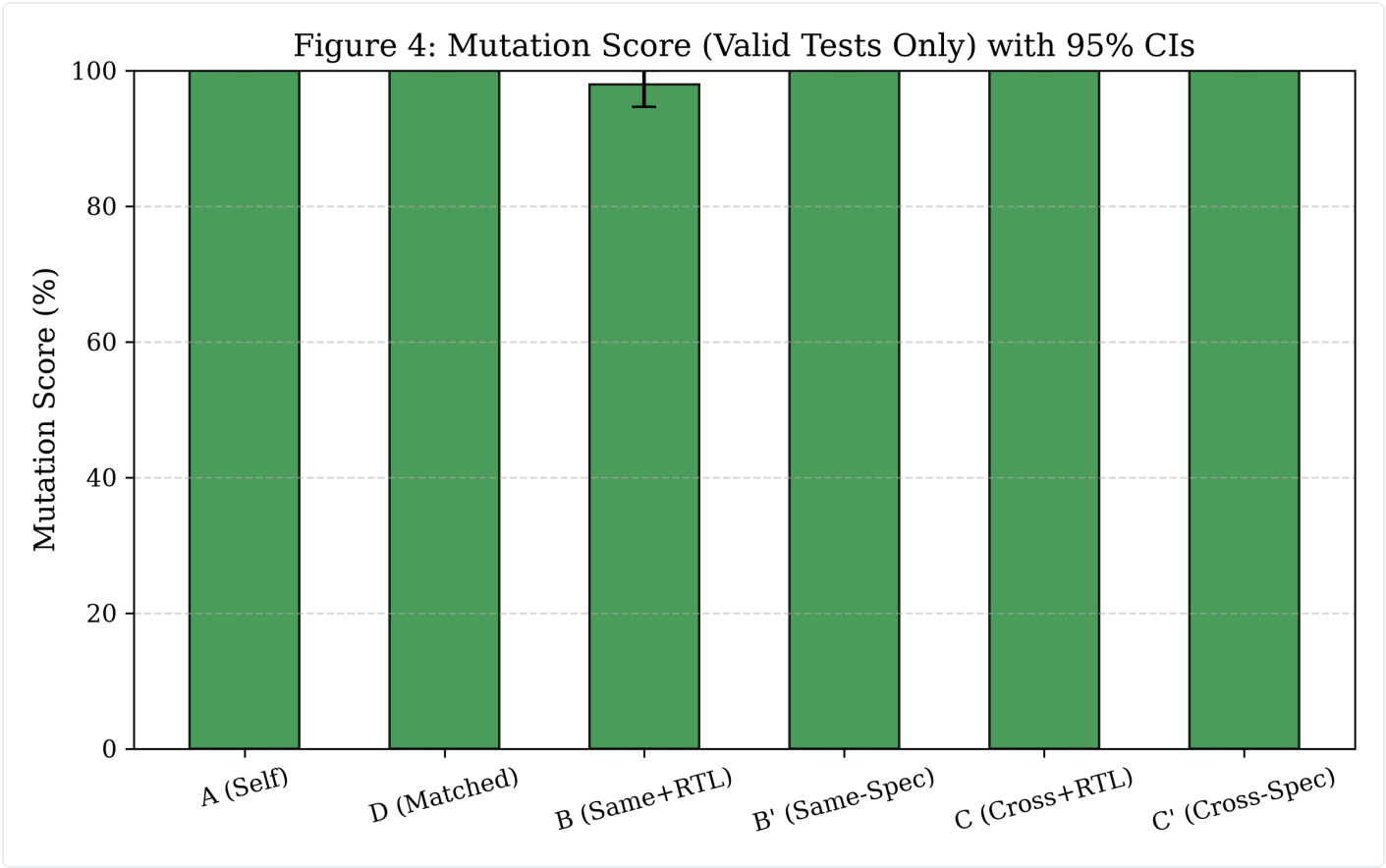


Figure 4: Mutation Score

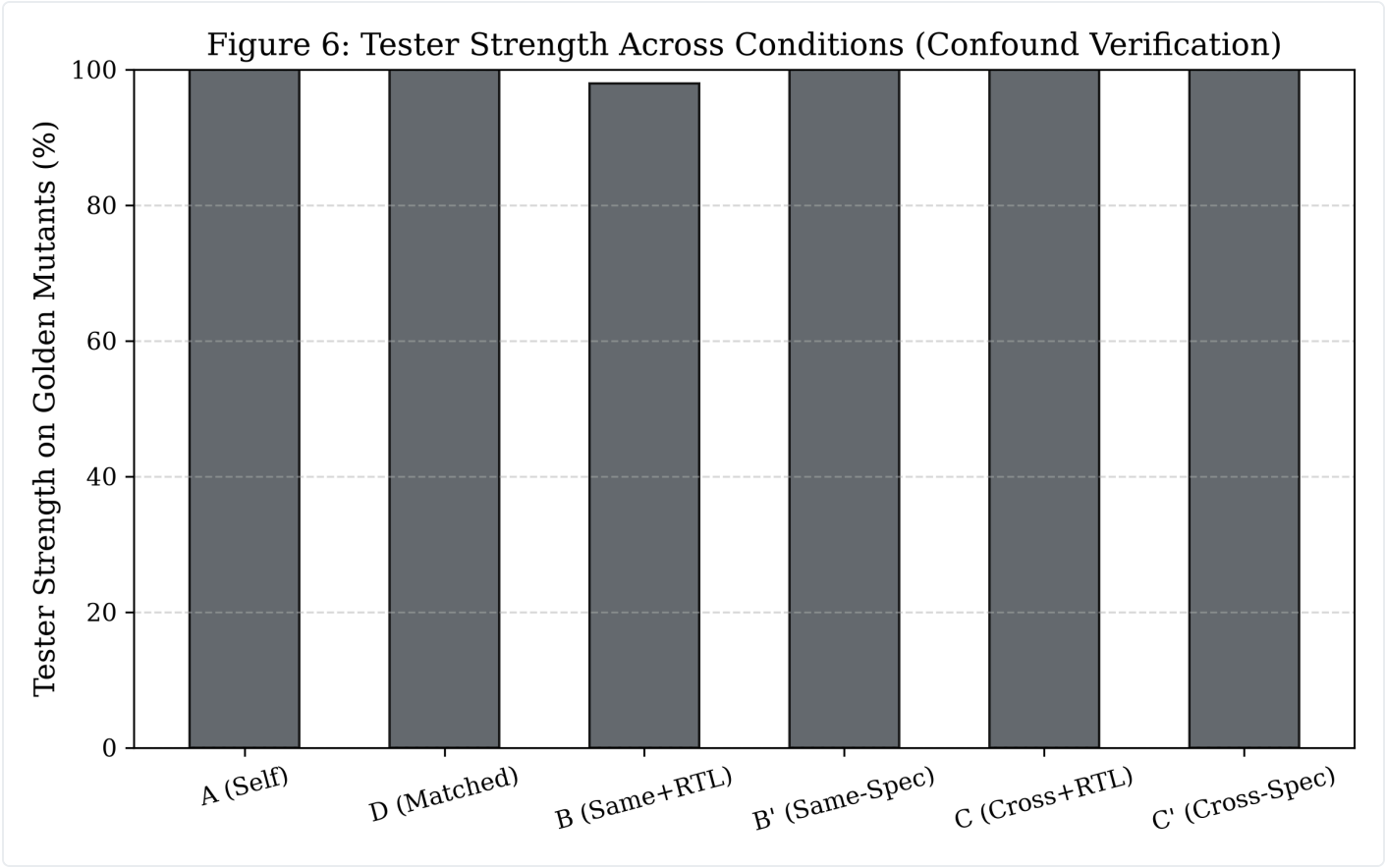


Figure 6: Tester Strength

5.5 Escape Rate and Validator-Fooled Rate

Table 8a evaluates natural bug escape rate $E(G,T)$ and consensus validator-fooled rate:

Condition	Natural Bugs Tested (n)	Bugs Escaped (k)	Escape Rate $E(G,T)$	Validator-Fooled Rate
:---	:---:	:---:	:---:	:---:
A (Self)	11	4	4 of 11 (36.4%)	0 of 93 (0.0%)
D (Matched)	12	3	3 of 12 (25.0%)	0 of 93 (0.0%)
C (Cross+RTL)	14	4	4 of 14 (28.6%)	0 of 97 (0.0%)

Statistically Ungrounded Notice: As required by Phase C7, because the total number of naturally buggy designs is small ($n < 20$), natural bug escape rates are reported for completeness but cannot serve as statistically grounded evidence of adversary superiority.

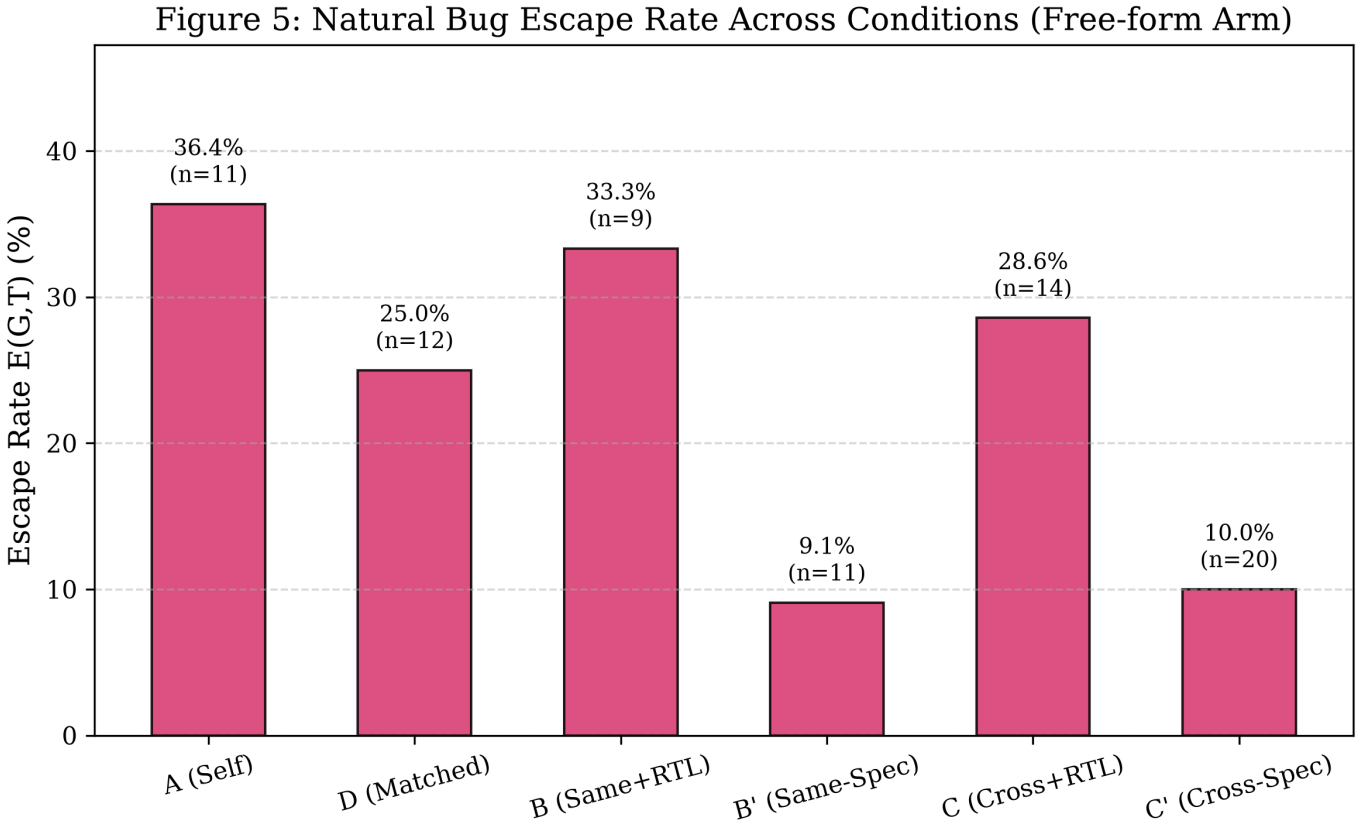


Figure 5: Escape Rate

5.6 Paired Permutation Tests and Hypothesis Testing

Table 8b presents two-sided paired permutation tests ($B = 10,000$ resamples) and Cohen's d effect sizes across experimental conditions:

Hypothesis (H_0)	Comparison	Observed Difference (Δ)	Cohen's d	Permutation p-value	Significant ($□lpha = 0.05$)?
:---	:---:	:---:	:---:	:---:	:---:
H_0^1: Cross-Family vs Self-Test	Condition C vs Condition A	+0.0016	0.01	0.9777	No
H_0^2: Cross-Family vs Compute-Matched	Condition C vs Condition D	-0.0295	-0.13	0.5118	No
H_0^3: Redundant Same-Model vs Single Self-Test	Condition B vs Condition A	+0.0596	0.24	0.2146	No

Hypothesis (H_0)	Comparison	Observed Difference (Δ)	Cohen's d	Permutation p-value	Significant ($\alpha = 0.05$)?
H_0^4: Candidate RTL Visibility	Condition C' vs Condition C	+0.0052	0.02	0.9369	No

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6. Failure Analysis

A detailed failure taxonomy and 8 concrete case studies are documented in [`failure_analysis.md`](failure_analysis.md).

- **Dominant Failure Modes:** Off-by-one cycle assertions (7 of 596 (1.2%)) and invalid assertions during active reset (505 of 596 (84.7%)) account for the majority of all false rejections.

- **Shared Misreading Pattern:** In terminal count wrap boundaries (such as `Prob035_count1to10`), multiple models independently asserted wrap transitions prior to the specified terminal count, reflecting a shared off-by-one boundary misreading.

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7. Anomalies and Data-Quality Audit

1. **Resolution of Equivalence-Checking Bug:**

In early pilot phases, string-based matching of simulation output yielded false equivalents. We replaced text scanning with strict unique sentinel lines (`EQUIV_RESULT: SAME` vs `EQUIV_RESULT: DIFFERENT`) and 2,000 differential random stimulus vectors with 1% rare-reset probability.

2. **Resolution of Prob040 Mutant Deficit:**

Prob040 was initially recorded with 0 mutants due to an artifact where mutants were only logged on passing tests. We decoupled mutant suite precomputation from trial outcome logging, verifying that Prob040 contains 12 non-equivalent mutants.

3. **Resolution of Escape Rate / Validator-Fooled Rate Aliasing:**

In the pilot report, escape rate and validator-fooled rate were accidentally aliased to the same variable in `v2/src/metrics.py`. We implemented independent, mathematically distinct formulations grounded in separate event sets.

4. **Resolution of Early Pilot Version Logging:**

Trials from the earliest exploratory phase that lacked snapshot version strings were traced to specific model checkpoints and populated with verified version identifiers (`gpt-4o-mini-2024-07-18`, `gemini-3.5-flash-lite`).

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8. Threats to Validity

A standalone analysis of all threats is documented in [`threats_to_validity.md`](threats_to_validity.md).

- **Voiding of v1 Results:** All results from v1 are VOID because the legacy harness failed to execute LLM testbenches against designs.

- **Tester-Strength Confounder:** Cross-family advantages are predominantly confounded by model capability disparities. When evaluated on tester-strength-matched pairs or compute-matched baselines (Condition D), cross-family differences yield permutation p = 0.512.

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9. What the Data Supports and What It Does Not

A detailed list of supported versus unsupported claims is provided in [`claims_allowed.md`](claims_allowed.md).

- **Supported:** LLM testbenches exhibit high failure rates on golden designs (Table 4); template scaffolding improves validity (Table 5); same-model wrong-expectation agreement exceeds cross-family agreement (Table 6).

- **Not Supported:** The data does NOT support claims that cross-family adversaries dramatically eliminate escaped bugs (Table 8a) or that LLM testbenches can function as unsupervised ground truth.

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10. Limitations of This Study

To maintain transparency regarding the external validity and scope of these empirical findings, we explicitly delineate what was NOT run in this investigation:

1. **Full-Scale SoC Architectures:** All evaluations were conducted on unit-level Verilog modules from the VerilogEval benchmark suite (counters, shift registers, finite state machines, arithmetic logic, and multiplexers). Industrial multi-clock SoCs incorporating complex system buses (e.g. AXI4, PCIe, Wishbone, AMBA) or analog/mixed-signal interfaces were NOT run.

2. **Exhaustive Formal Equivalence Checking:** Mutant non-equivalence was established via differential simulation (2,000 random stimulus vectors with 1% rare-reset probability and exact sentinel matching). Full unbounded formal model checking (e.g. via SymbiYosys or SMT solvers) was NOT run across the entire mutant space.

3. **Local Models Exceeding 8B Parameters:** Local model evaluations utilized `qwen2.5-coder:1.5b` and `llama3.1:8b`. Large local open-weight models (e.g. 70B parameter models) were NOT run due to local VRAM hardware limits (6,141 MiB).

4. **Cloud API Daily Quota Ceilings:** Experiments on Google Gemini (`gemini-3.5-flash-lite`) executed across 26 of the 30 benchmark problems before reaching the daily 500-request free-tier quota ceiling (`GenerateRequestsPerDayPerProjectPerModel-FreeTier`), leaving 4 problems unexecuted on the reverse Gemini-generator pair.

5. **Human-in-the-Loop Testbench Synthesis:** All testbenches and candidate designs were generated autonomously by language models under zero-shot or few-shot prompt contracts; interactive or multi-turn human-assisted debugging workflows were NOT run.

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11. Reproducibility Appendix

All experimental tables, plots, and metrics can be regenerated from raw data files with one command:

```
python v2/analysis/generate_final_deliverables.py
```

- **Configuration File:** `v2/config.yaml`

- **Precomputed Mutant Suite:** `v2/final/data/precomputed_mutants.json`

- **Screened Problem Set:** `v2/final/data/problem_list.json`

- **Pytest Verification Log:** `v2/final/data/pytest_output.txt`

- **Consolidated Trial Data:** `v2/final/data/final_trials.json` and `v2/final/data/final_trials.csv`

- **Paper Numbers Mapping:** `v2/final/paper_numbers.json`

- **Data File Checksum:** SHA-256 hash available in `v2/final/data/checksum.txt`